

BENCHMARK OVERVIEW

Ryzen AI MAX+ 395 Before vs After V5

NP1 parameter-change comparison · Context + Power controlled rerun

1) Benchmark scope

Models	Gemma 4 26B A4B / GPT-OSS 20B
Inputs	1K / 2K / 4K / 8K / 16K / 32K
Output / Trials	300 tokens / 3 runs
Runtime	llama.cpp build 10615 · Vulkan

2) Comparison modes

395 Before	Reference result set
395 V5 After	Comparison result set
Δ convention	Positive = 395 V5 After faster
Model SHA256	Matched for both models

3) Metrics in this report

TTFT (ms)	Request start → first token
TPOT (ms/token)	Decode time per generated token
Aggregate throughput	Total output capacity (tok/s)
Normalized throughput	Per-user capacity (tok/s/user)

4) Reading guide

TTFT / TPOT	Lower is better
Throughput	Higher is better
NP1	Aggregate = Normalized
Interpretation	Observed benchmark deltas

METHOD NOTE Before vs After changes Context and Power together: results show combined observed effect, not each factor independently.

Parameter	395 Before	395 V5 After
Context	65,536	dynamic 4K-34.8K
Threads	32	32
Power mode	Silent	High Performance + OEM Performance/Turbo
Driver	32.0.31035.1003	32.0.31035.1003
Vulkan memory	24,395 MiB	24,395 MiB

TTFT

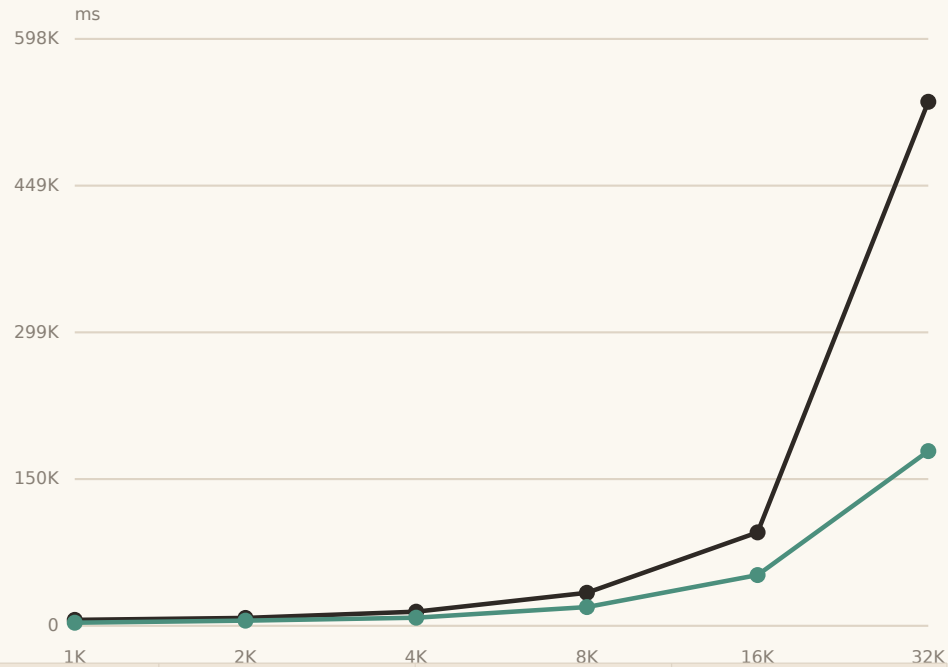
Time to first token (ms). Lower is better.

395 Before 395 V5 After

FORMULA $\Delta (\%) = (\text{Reference} - \text{Comparison}) / \text{Reference} \times 100$
Positive means 395 V5 After measured faster

Gemma 4 26B A4B

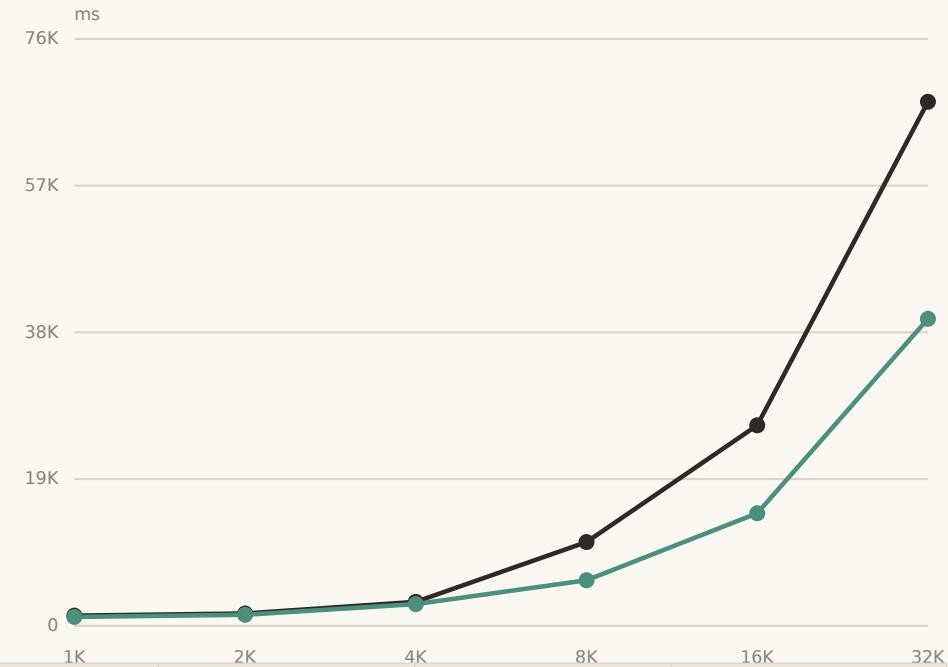
NP1 | Output 300



Input	395 Before	395 V5 After	After Δ
1K	6,037	3,148	+47.9%
2K	8,018	5,256	+34.5%
4K	14,394	8,259	+42.6%
8K	33,654	19,095	+43.3%
16K	95,240	51,734	+45.7%
32K	534,246	178,107	+66.7%

GPT-OSS 20B

NP1 | Output 300



Input	395 Before	395 V5 After	After Δ
1K	1,321	1,154	+12.7%
2K	1,599	1,430	+10.6%
4K	3,110	2,818	+9.4%
8K	10,874	5,912	+45.6%
16K	26,037	14,615	+43.9%
32K	68,043	39,859	+41.4%

KEY TAKEAWAY At 32K, V5 change: Gemma +66.7%; GPT-OSS +41.4%. Positive means the V5 run is better.

TPOT

TPOT

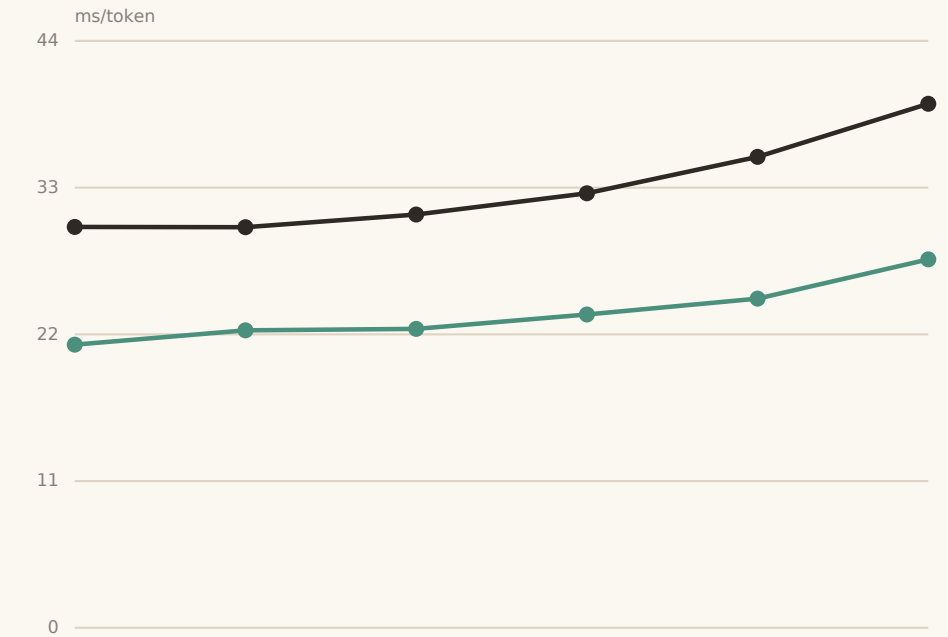
Median time per output token (ms/token). Lower is better.

395 Before 395 V5 After

FORMULA $\Delta (\%) = (\text{Reference} - \text{Comparison}) / \text{Reference} \times 100$
Positive means 395 V5 After measured faster

Gemma 4 26B A4B

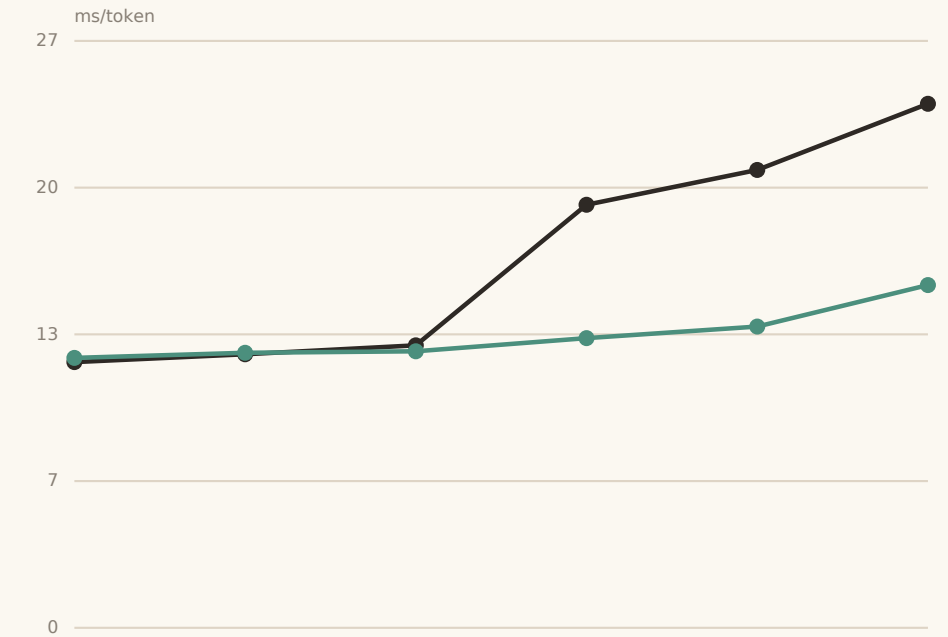
NP1 | Output 300



Input	395 Before	395 V5 After	After Δ
1K	30.37	21.45	+29.4%
2K	30.35	22.54	+25.7%
4K	31.31	22.65	+27.7%
8K	32.92	23.74	+27.9%
16K	35.68	24.94	+30.1%
32K	39.70	27.91	+29.7%

GPT-OSS 20B

NP1 | Output 300



Input	395 Before	395 V5 After	After Δ
1K	12.21	12.40	-1.6%
2K	12.57	12.64	-0.5%
4K	12.98	12.71	+2.1%
8K	19.44	13.31	+31.5%
16K	21.04	13.84	+34.2%
32K	24.08	15.75	+34.6%

KEY TAKEAWAY At 32K, V5 change: Gemma +29.7%; GPT-OSS +34.6%. Positive means the V5 run is better.

THROUGHPUT

Throughput (Aggregate)

Total server output token throughput (tok/s). Higher is better.

395 Before 395 V5 After

FORMULA $\Delta (\%) = (\text{Comparison} - \text{Reference}) / \text{Reference} \times 100$
Positive means 395 V5 After measured faster

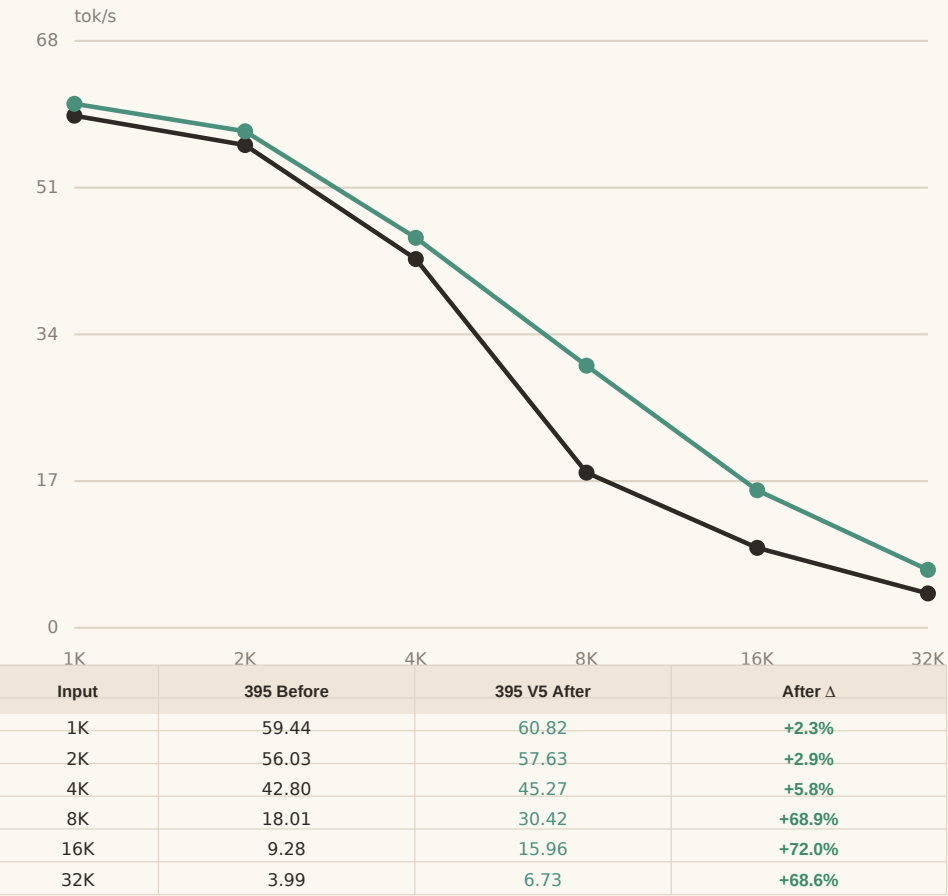
Gemma 4 26B A4B

NP1 | Output 300



GPT-OSS 20B

NP1 | Output 300



KEY TAKEAWAY At 32K, V5 change: Gemma +192.5%; GPT-OSS +68.6%. Positive means the V5 run is better.

THROUGHPUT

Throughput (Normalized)

Aggregate throughput divided by NP (tok/s/user). Higher is better.

395 Before 395 V5 After

FORMULA $\Delta (\%) = (\text{Comparison} - \text{Reference}) / \text{Reference} \times 100$
Positive means 395 V5 After measured faster

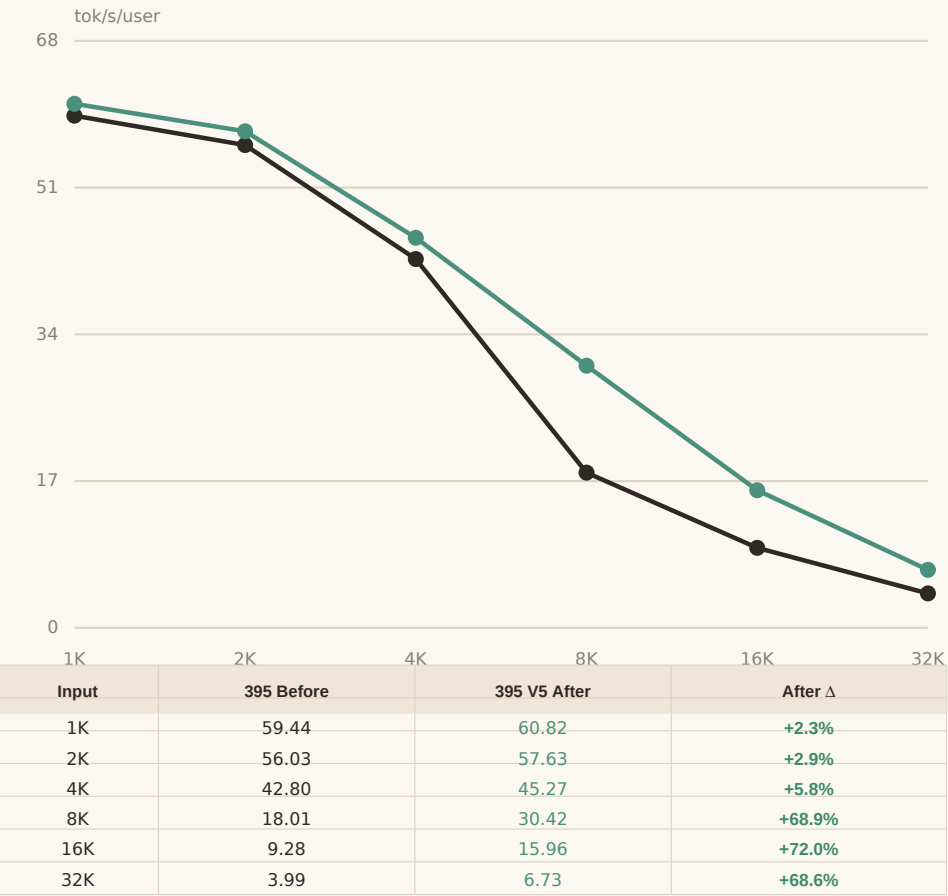
Gemma 4 26B A4B

NP1 | Output 300



GPT-OSS 20B

NP1 | Output 300



KEY TAKEAWAY At 32K, V5 change: Gemma +192.5%; GPT-OSS +68.6%. Positive means the V5 run is better.

SUMMARY

Observed impact on latency, decode efficiency, and throughput

AI MAX+ 395 before → V5 after · NP1 · Output 300 · focus on 32K

TTFT

ms

Gemma 4 26B	534,246 → 178,107	+66.7%
GPT-OSS 20B	68,043 → 39,859	+41.4%

TPOT

ms/token

Gemma 4 26B	39.70 → 27.91	+29.7%
GPT-OSS 20B	24.08 → 15.75	+34.6%

Throughput (Aggregate)

tok/s

Gemma 4 26B	0.55 → 1.61	+192.5%
GPT-OSS 20B	3.99 → 6.73	+68.6%

Throughput (Normalized)

tok/s/user

Gemma 4 26B	0.55 → 1.61	+192.5%
GPT-OSS 20B	3.99 → 6.73	+68.6%

Key interpretation

- The V5 rerun substantially improves long-context performance on the same AI MAX+ 395.
- Gemma 32K: TTFT improves 66.7%, TPOT 29.7%, and throughput 192.5% versus the earlier 395 run.
- GPT-OSS 32K: TTFT improves 41.4%, TPOT 34.6%, and throughput 68.6%.
- The changed variables are Context policy and Power mode; Driver and Vulkan/VGM state were intentionally left unchanged.